

An Evidence-Hardening Funnel for Transcriptomic Prognostic Claims in Clear Cell Renal Cell Carcinoma

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Abstract—Public transcriptomic studies often treat large tumor–normal expression changes as candidate prognostic biomarkers without testing whether the associations are stable, externally reproducible, or driven by tissue composition. We developed HYDRA-ccRCC, a reproducible evidence-hardening workflow for clear cell renal cell carcinoma. TCGA-KIRC STAR-count RNA sequencing was used for discovery; GSE40435 and GSE53757 tested tumor–normal replication; GSE29609 and E-MTAB-1980 provided independent survival checks. Continuous-expression Cox models adjusted for available clinical covariates. The analysis identified 3,304 reproducible differentially expressed genes, 519 strict TCGA-derived prognostic candidates, and a frozen 24-candidate high-confidence set. In E-MTAB-1980, 23 candidates mapped, 20 retained the TCGA hazard direction, and eight passed prespecified direction, unadjusted and adjusted false-discovery-rate, and proportional-hazards gates: DDC, CRYL1, ACADM, TEK, EMCN, PODXL, FUT6, and HIBCH. GSE29609 was not broadly supportive. In 100 patient-level bootstraps, eight frozen candidates were selected in at least half of repeats, but none reached 80%. Conditional out-of-bag concordance changes were positive on average for 22 candidates, while every percentile interval crossed zero. All 24 candidates retained direction and FDR support after adjustment for published consensus tumor purity. Normal-tissue single-cell data identified vascular-dominant and immune-dominant candidates, reinforcing composition cautions. The most convergent metabolic hypotheses were ACADM, CRYL1, and DDC, whereas several externally supported genes remained likely compartment markers. HYDRA-ccRCC prioritizes candidate prognostic associations while exposing instability, cohort disagreement, and composition dependence; it does not establish a validated biomarker panel.

Index Terms—clear cell renal cell carcinoma, TCGA, GEO, transcriptomics, survival analysis, external validation, selection stability

I. INTRODUCTION

Clear cell renal cell carcinoma (ccRCC) is shaped by VHL loss, hypoxia-inducible signaling, angiogenesis, altered metabolism, and a heterogeneous tumor microenvironment [1], [2]. These processes produce broad transcriptomic differences between tumor and normal kidney. A large differential-expression effect, however, can reflect tumor identity, loss of normal nephron compartments, immune or vascular admixture, or technical context rather than stable prognostic information.

HYDRA-ccRCC tests a narrow question: among genes reproducibly dysregulated across independent ccRCC expression cohorts, which have survival associations that remain credible after clinical adjustment, proportional-hazards testing, sensitivity analysis, resampling, independent outcome testing,

and cell-source triangulation? The contribution is an auditable evidence funnel rather than a new prediction algorithm or a fixed multigene panel.

II. METHODS

A. Cohorts and Frozen Analysis Design

TCGA-KIRC public, de-identified data were obtained from the Genomic Data Commons through TCGAAbiolinks [3]. Discovery used 541 primary tumors and 72 solid-tissue normal samples. GSE40435 (101 tumor and 101 normal samples) and GSE53757 (72 tumor and 72 normal samples) were obtained from the Gene Expression Omnibus [4] for tumor–normal replication.

The 24 high-confidence candidates were frozen before v2 external testing. GSE29609, with 39 tumors and 17 deaths, was retained as a small exploratory survival-direction check. E-MTAB-1980, with 101 patients and 23 deaths, was the primary frozen external survival test. No external result changed the TCGA candidate definition or ranking.

B. Differential Expression and Reproducibility

TCGA raw unstranded STAR counts were analyzed with DESeq2 [5]. Genes required at least 10 counts in at least 10 samples. Significance required Benjamini–Hochberg adjusted $p < 0.05$ and absolute $\log_2$ fold change of at least 1.

The GEO cohorts were analyzed with limma [6]. GSE40435 used parsed patient-pair identifiers; GSE53757 used the documented alternating tumor–normal sample order. Probe expression was collapsed to gene symbols by mean expression. A reproducible DEG required TCGA significance, the same tumor–normal direction in both GEO cohorts, and nominal $p < 0.05$ in at least one GEO cohort.

C. TCGA Survival Selection

Overall survival models used TCGA primary tumors and standardized continuous expression. The main Cox model [7] adjusted for age, sex, pathologic stage, and collapsed grade among complete cases. Stage-complete and grade-complete sensitivity models were fit separately. High-confidence candidates required main-model FDR < 0.01 , proportional-hazards $p \geq 0.05$, absolute log hazard ratio at least $\log(1.5)$, non-trivial GEO effect support, and concordant nominal support in both sensitivity models.

D. External Survival Testing

External expression values were standardized within cohort. GSE29609 tested unadjusted continuous-expression Cox associations and retained missing and contradictory results. E-MTAB-1980 tested all frozen candidates with an unadjusted primary model and a limited secondary model adjusting for age, high T stage, and metastatic status. Candidate-wise multiplicity was controlled with Benjamini–Hochberg FDR. Strict external support required TCGA-concordant direction, FDR support in both external models, and proportional-hazards support in both models.

E. Selection Stability and Held-Out Clinical Increment

Selection stability used 20 event-stratified 80% TCGA subsamples. Each repeat reran the complete survival-selection screen across all 3,304 eligible reproducible DEGs, rather than refitting only the frozen candidates. We recorded candidate selection frequency, median effect, direction agreement, and the number selected per repeat.

Predictive increment used 20 repeats of five-fold event-stratified cross-validation. For each candidate, a clinical-only Cox model and a clinical-plus-one-gene model were trained within each fold and evaluated on held-out patients. Out-of-fold predictions were pooled within each repeat, and the concordance difference was summarized across repeats. These repeated-split distributions are internal uncertainty summaries, not independent confidence intervals or evidence of clinical utility.

Following independent statistical feedback, a second analysis used 100 event-stratified patient bootstraps. Each repeat reran the complete 3,304-gene survival-selection screen in the bootstrap sample. Frozen candidates that passed the complete screen were then refit with the clinical model and evaluated only in patients omitted from that bootstrap. Selection frequency, bootstrap effect intervals, and conditional out-of-bag concordance changes were retained. This analysis starts from the independently replicated DEG set and does not rerun upstream DESeq2 or GEO processing.

F. Composition and Cell-Source Checks

Clinical-plus-gene models were compared with models that additionally included proximal-tubule, endothelial, immune, and stromal expression-derived marker scores. Candidate-specific leave-one-out scores reduced direct circularity when a candidate overlapped a marker set. These scores are composition screens, not direct tumor-purity estimates.

Published TCGA consensus purity estimates from Aran et al. [8] were matched by aliquot identifier. Each frozen candidate was refit with standardized purity, age, sex, stage, and grade. The analysis recorded direction agreement, false-discovery rate, proportional-hazards support, and attenuation relative to the original clinical model.

Human Protein Atlas v25.1 normal-tissue single-cell expression was mapped to renal epithelial, vascular, immune, stromal, and other compartments using a prespecified rule set. The analysis recorded the dominant mapped compartment

and top cell types for every frozen candidate. Because the source contains normal tissues rather than ccRCC tumor cells, it supports cell-source triangulation but cannot establish tumor-cell expression or mechanism.

G. Reproducibility and Validation

The scripted pipeline records public source URLs, access dates, input checksums, random seeds, package versions, and output checksums. Automated tests check required columns, expected candidate coverage, cohort counts, monotonic funnel behavior, and impossible values. The manuscript is compiled separately from the analysis pipeline.

III. RESULTS

A. The Evidence Funnel Compresses Broad Dysregulation

TCGA analysis identified 14,337 significant Ensembl-level genes and 8,852 genes after symbol mapping. GSE40435 and GSE53757 identified 1,076 and 3,139 significant genes, respectively. Cross-cohort filtering retained 3,304 reproducible DEGs. Of these, 1,176 passed main Cox FDR, 1,122 passed proportional-hazards testing, 1,063 passed sensitivity requirements, 519 met the strict definition, and 24 met the high-confidence definition.

HYDRA-ccRCC Evidence Funnel

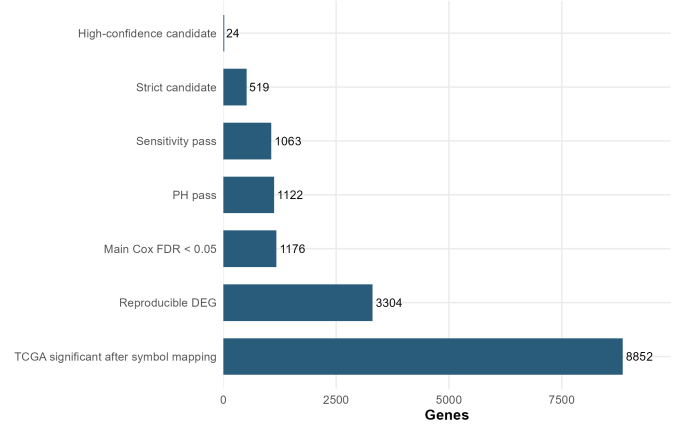


Fig. 1. Evidence funnel from TCGA tumor–normal differential expression to the frozen high-confidence candidate set.

TABLE I
DISCOVERY EVIDENCE FUNNEL

Metric	Count
TCGA tumor / normal samples	541 / 72
TCGA significant mapped genes	8,852
Reproducible DEGs	3,304
Main Cox FDR pass	1,176
Proportional-hazards pass	1,122
Sensitivity pass	1,063
Strict candidates	519
High-confidence candidates	24

Differential-expression magnitude and adjusted survival effect were not equivalent. Many protective candidates were

lower in tumor, consistent with retained renal epithelial or metabolic differentiation, while several risk-direction genes were compatible with immune, inflammatory, vascular, or stromal states.

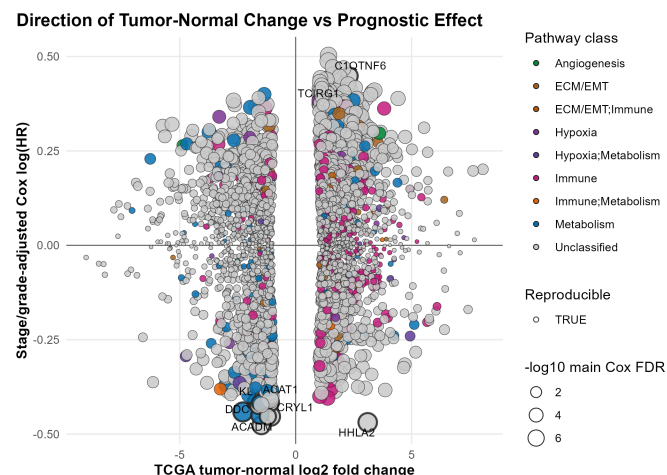


Fig. 2. Directional relationship between tumor-normal expression change and adjusted TCGA survival effect.

B. Independent Survival Cohorts Give Unequal Support

GSE29609 tested 22 platform-present candidates. Four retained the TCGA hazard direction, none had same-direction nominal support, and DDC, ACAT1, CLCN5, and TCIRG1 had nominal opposite-direction associations. Its 17 events and distinct measurement platform limit gene-specific inference, but the result rejects a claim of universal cross-platform validation.

E-MTAB-1980 mapped 23 candidates. Twenty retained the TCGA direction, 14 had same-direction unadjusted FDR support, 12 had adjusted FDR support, and eight passed the complete external gate. No candidate had a nominally significant opposite-direction association.

TABLE II
EXTERNAL SURVIVAL EVIDENCE

Metric	GSE29609	E-MTAB-1980
Samples / events	39 / 17	101 / 23
Candidates mapped	22	23
Same TCGA direction	4	20
Same-direction nominal	0	14
Strict external support	—	8
Opposite-direction nominal	4	0

The strict E-MTAB-1980 set was DDC, CRYL1, ACADM, TEK, EMCN, PODXL, FUT6, and HIBCH. Statistical replication alone did not make these genes biologically equivalent: TEK and EMCN were vascular signals, PODXL was compartment-sensitive, and FUT6 and HIBCH had limited candidate-specific biological support.

C. Top Candidates Are Stable but the Threshold-Defined Set Varies

The complete selection screen chose between 3 and 129 genes per resample, with a mean of 46.9. Six frozen candidates were selected in at least 80% of repeats: PANK1 (100%), ACADM (95%), DDC (90%), CLCN5 (85%), CRYL1 (85%), and CYFIP2 (85%). Eighteen of 24 were selected in at least half of repeats, and all 24 retained their full-fit hazard direction by median resampled effect. Thus, several leading associations were stable, but the total count passing hard thresholds was sensitive to sampling variation.

D. Single Genes Add Small Internal Discrimination

Across repeated held-out splits, the clinical-only model had mean concordance 0.745. All 24 candidates produced a positive mean concordance change when added individually; 23 had a positive empirical 2.5th percentile, while FHOD1 crossed zero. The largest mean increments were CYFIP2 (+0.0220), PANK1 (+0.0218), CRYL1 (+0.0206), ACADM (+0.0181), HHLA2 (+0.0176), and CLCN5 (+0.0175). These small internal gains do not define a multigene model or show clinical usefulness.

E. Patient-Level Bootstrap Tempers the Performance Claim

Across 100 patient-level bootstraps, the complete survival-selection screen chose between 4 and 207 genes, with a mean of 65.4. Eight frozen candidates were selected in at least half of repeats: PANK1 (66%), CLCN5 (60%), ACADM (59%), CRYL1 (56%), TCIRG1 (55%), GRAMD1A (51%), LRBA (51%), and HIBCH (50%). None reached 80%. All 24 retained their full-fit hazard direction, and all percentile intervals for the adjusted log hazard ratio excluded zero.

Out-of-bag sets averaged 190.6 patients and 62.9 deaths. Conditional on passing the selection screen, 22 candidates had a positive mean out-of-bag concordance change; DBT and FHOD1 were negative on average. Every 95% percentile interval crossed zero. The largest positive means were CYFIP2 (+0.0214), PANK1 (+0.0167), HHLA2 (+0.0152), CRYL1 (+0.0143), and ACADM (+0.0119). Thus, the bootstrap supports recurring association more strongly than reliable incremental prediction.

F. Cell Source Changes the Candidate Hierarchy

All 24 candidates retained TCGA hazard direction after bulk marker-score adjustment, but only 16 remained significant at $FDR < 0.05$. KL and DDC attenuated, suggesting sensitivity to renal epithelial differentiation or composition.

Consensus purity estimates matched 516 complete-case TCGA tumors with 170 deaths. After direct purity adjustment, all 24 candidates retained their original hazard direction and remained significant at $FDR < 0.05$. The largest relative attenuation in absolute log hazard ratio was 8.9%, for GRAMD1A. Measured consensus purity therefore did not explain the TCGA candidate associations.

Human Protein Atlas data mapped all 24 genes. TEK and EMCN were vascular-dominant; CYFIP2, GRAMD1A, and

TCIRG1 were immune-dominant. Several remaining genes had renal epithelial expression even when their global maximum occurred in another normal-tissue cell type. These observations support compartment-aware interpretation but do not establish ccRCC tumor-cell origin.

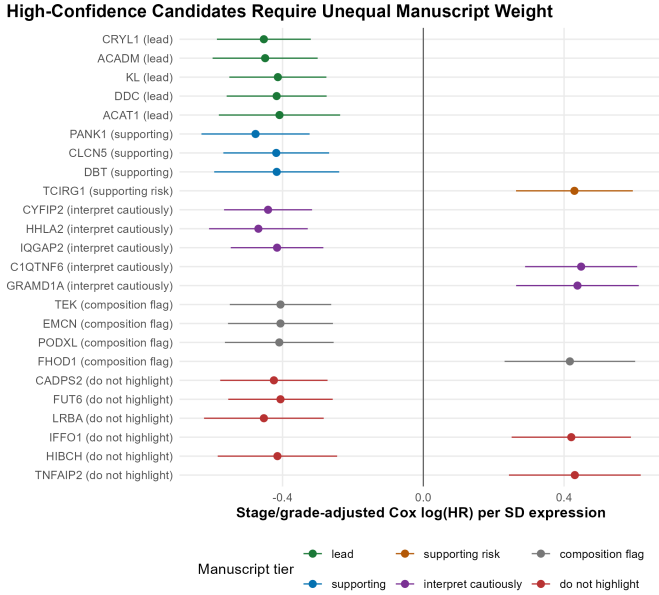


Fig. 3. Adjusted TCGA associations for the frozen 24-candidate set, grouped by manuscript role.

G. Integrated Interpretation Prioritizes Three Metabolic Hypotheses

ACADM, CRYL1, and DDC had the strongest convergence of TCGA association, selection stability, positive held-out increment, and strict E-MTAB-1980 support. Their direction is consistent with loss or retention of renal epithelial metabolic differentiation. PANK1, CLCN5, and CYFIP2 were internally stable but did not pass the strict external gate. KL and ACAT1 remained biologically plausible TCGA hypotheses without strict E-MTAB-1980 support.

The findings do not support a generic hypoxia-only explanation. VHL/HIF biology can reshape oxidative metabolism and tissue state, but the candidate evidence more directly reflects metabolic differentiation plus immune and vascular composition. The results prioritize experiments and literature review; they do not establish a mechanism or therapeutic target.

IV. DISCUSSION

HYDRA-ccRCC shows why prognostic transcriptomic claims need several distinct tests. Expression replication reduced broad TCGA dysregulation, survival selection reduced it further, and external testing separated a supported subset from candidates that remained cohort-dependent. Full-screen resampling showed that the strongest candidates recurred even though the number crossing hard thresholds varied substantially. The patient-level bootstrap also showed that small

repeated-cross-validation gains did not translate into precise out-of-bag performance increments.

The two external survival cohorts should be interpreted together. GSE29609 supplied a useful contradictory result, but its small event count and microarray coverage limit power. E-MTAB-1980 provided stronger directional and multiplicity-controlled support, but its 23 deaths still constrain adjustment and precision. Eight strict external hits are evidence for candidate-level replication, not evidence for a deployable panel.

The convergent metabolic trio—ACADM, CRYL1, and DDC—is the most defensible next-stage focus. DDC’s opposite nominal direction in GSE29609 prevents a simple replication narrative, while its strong E-MTAB-1980 result and resampling stability justify further study. Vascular and immune candidates demonstrate why statistical validation must be paired with cell-source analysis.

V. LIMITATIONS

This retrospective study uses public bulk transcriptomic data. TCGA supplies discovery, selection, and repeated internal evaluation. The patient bootstrap reduces apparent optimism but remains internal, starts from the fixed cross-cohort DEG set, and reports out-of-bag performance conditional on selection. E-MTAB-1980 has only 23 deaths and permits limited covariate adjustment; GSE29609 has 17 deaths, different technology, and incomplete candidate coverage. Overall survival is affected by treatment, comorbidity, and follow-up differences.

Adjacent normal kidney is not equivalent to healthy kidney. Consensus purity is inferred from several molecular and pathology estimates and is not direct histopathologic measurement in this analysis. HPA v25.1 contains normal-tissue rather than ccRCC tumor single-cell data. Bulk RNA cannot separate tumor-cell-intrinsic expression from nephron retention, immune infiltration, endothelial content, stromal admixture, or necrosis. RNA abundance also does not demonstrate protein effect, causation, mechanism, or therapeutic relevance.

VI. CONCLUSION

The completed v2 workflow narrows thousands of reproducible tumor-normal changes to a frozen candidate set, tests that set in two external survival cohorts, quantifies full-screen selection stability and patient-level bootstrap uncertainty, evaluates out-of-bag clinical increment, adjusts directly for consensus tumor purity, and triangulates normal-tissue cell source. ACADM, CRYL1, and DDC emerge as the strongest integrated metabolic hypotheses, while several externally supported genes remain composition-sensitive. The appropriate output is a prioritized set of candidate prognostic associations and a transparent record of where the evidence remains fragile, not a validated 24-gene biomarker panel.

DATA AVAILABILITY

All source data are public and identified by TCGA-KIRC, GSE40435, GSE53757, GSE29609, E-MTAB-1980, Human

Protein Atlas v25.1, and the Aran et al. TCGA purity supplement. Download URLs, access dates, and checksums are recorded in `./results/tables/source_provenance.csv`; generated tables and figures are stored under `./results/`; and the one-command entry point is `./run_pipeline.ps1`.

ETHICS DECLARATION

This secondary analysis used public, de-identified datasets and involved no new recruitment, intervention, or access to identifiable private information. Final competition or publication requirements should be confirmed with the relevant review body.

AUTHOR CONTRIBUTIONS

The student researcher is responsible for the project question, methodological decisions, verification of the analysis, interpretation, and final writing. Contributions must be reviewed and updated before submission to match the student's documented work.

FUNDING

No external funding was reported for this development-stage project.

CONFLICT OF INTEREST

The author reports no conflict of interest.

AI ASSISTANCE DISCLOSURE

OpenAI Codex assisted with code auditing, implementation support, debugging, reproducibility checks, and revision of this internal development manuscript. The student remains responsible for understanding and independently verifying the code, statistical decisions, citations, and scientific claims. AI-generated prose must not be copied directly into IRIS or ISEF submission materials.

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